Solution Architecture

Layered backend architecture

Project: AI WeatherWise API   Track: AI-ML & GEN-AI Track

Prepared: 07 October 2026   Team ID: Not provided

Maximum Marks: 5

Team Members & Assigned Roles

Team Member	Role
Jai Bharath A	Team Lead & Lead Backend/API Architect
Mukesh Balaji	Generative AI & Prompt Engineering Engineer
Yadagiri Mani	Database & Data Engineering Specialist
Deepak G	Security, Testing & Reliability Engineer
Naveen	API Integration, Documentation & Demonstration Engineer

Architecture Component Table

Layer	Components	Responsibility
Client	Thunder Client / Web / Mobile	Sends HTTP requests and displays JSON.
API / Routing	Express routes + middleware	Routes requests, validates auth and invokes controllers.
Controller	Auth / Weather / Location / AI controllers	Orchestrates request-response behavior.
Service	weatherService.js / aiService.js	External API calls, normalization, AI prompts and fallbacks.
Data	Mongoose models + MongoDB Atlas	Persistent users and favorite locations.
External AI / Data	OpenWeatherMap + Gemini 2.5 Flash	Live observations and generative intelligence.

Request Sequence Example

Step	Component	Action
1	GET /api/weather/Chennai	Route receives city
2	weatherController	Calls weatherService
3	weatherService	Calls OpenWeatherMap
4	Failure?	Uses deterministic mock weather
5	Response	Normalized weather JSON returned

AI Sequence

- Client sends weather context to the AI endpoint.
- AI service constructs a constrained prompt emphasizing current metrics and practical advice.
- Gemini 2.5 Flash generates the response.
- The service parses/normalizes the response.
- If Gemini fails or quota is reached, deterministic rules produce a usable recommendation.

Security Boundary

Authentication middleware sits before protected profile and location routes. MongoDB queries include the authenticated user identity so a valid token does not automatically grant access to another user's saved locations.